

FitBuddy

Phase 3: Project Design Phase

Smart Fitness and Wellness Application

1. System Architecture

FitBuddy follows a three-tier architecture: the Presentation Layer (HTML, CSS, JavaScript) handles user interaction; the Application Layer (Python) contains business logic and the recommendation engine; the Data Layer (MySQL) stores all records.

Layer	Technology	Responsibility
Presentation	HTML, CSS, JavaScript	Forms, dashboard, charts, navigation
Application	Python	Authentication, recommendation logic, calculations, APIs
Data	MySQL	Users, goals, workouts, nutrition, activity logs

2. Database Design

Table	Key Fields
users	user_id (PK), name, email, password_hash, age, gender, height, weight, activity_level
goals	goal_id (PK), user_id (FK), goal_type, target_weight, start_date, end_date
workouts	workout_id (PK), name, category, difficulty, duration, calories_burned
nutrition	food_id (PK), food_name, calories, protein, carbs, fats
activity_log	log_id (PK), user_id (FK), workout_id (FK), date, duration, calories
progress	progress_id (PK), user_id (FK), date, weight, bmi

Relationships: one user has many goals, activity logs, and progress records; one workout can appear in many activity logs.

3. Data Flow

- User registers/logs in and enters profile details
- Backend validates and stores data in MySQL
- Recommendation engine reads profile and goal, selects matching workouts and tips
- User follows workout and records the activity

- Progress module aggregates logs and displays charts

4. Recommendation Logic (Overview)

- Compute BMI = weight (kg) / height (m) squared
- Map goal + activity level + BMI category to a workout difficulty and category
- Filter the workouts table and rank by suitability
- Suggest nutrition guidance matching the goal (deficit, surplus, or maintenance)

5. UI Design

Screen	Main Elements
Login / Register	Email, password, sign-up form
Profile Setup	Age, weight, height, goal, activity level
Dashboard	Today's plan, calories, weekly progress chart
Workouts	Recommended list with details and start button
Nutrition	Meal tips and calorie information
Activity Log	Form to record and view past activities
Progress	Weight/BMI trend graphs and summary

6. Security Design

Passwords are hashed, inputs are validated on client and server, SQL queries are parameterized to prevent injection, and sessions are managed securely.